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A middle Paleocene paleomagnetic pole from the Nacimiento Formation, San Juan Basin, New Mexico

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ABSTRACT

Paleomagnetic data from 104 sites (3 samples per site) in terrestrial sedimentary rocks of the Nacimiento Formation in the San Juan Basin were analyzed to determine a paleomagnetic pole for middle Paleocene time. Alternating-field demagnetization to 300-Oe peak field was sufficient to erase secondary viscous components. The mean direction of magnetization (inclination = 51.3°, declination = 343.9°) was calculated by averaging the site mean directions for the 42 normal polarity sites along with the antipoles of the site means of the 62 reversed polarity sites. The middle Paleocene pole position is at lat 75.9°N, long 147.7°E (δp = 2.7°, δm = 4.1°). The revised apparent polar wander path for North America shows a sharp turn for Late Cretaceous time that correlates with the onset of the Laramide orogeny.

INTRODUCTION

Despite over 20 years of paleomagnetic research, many intervals of the apparent polar wander path for North America are as yet poorly determined. It is particularly surprising that the details of apparent polar wander during early Tertiary time are not known. Although McElhinny (1973) listed nine early Tertiary pole positions available for North America, few of these results are based on a significant number of sampling sites. As discussed by Shive and others (1977), there are only three reliable Eocene results, and all of these have been determined within the past three years. All of these reliable Eocene results yield pole positions that are very near the present rotation axis. However, the Cretaceous pole position for North America is quite well determined (McElhinny, 1973) and has a location of lat 64°N, long 187°E with A_{95} of 6°. A large amount of apparent polar wander must have occurred between Late Cretaceous and Eocene time. There are no paleomagnetic pole position results for rocks of Paleocene age in North America (excluding Greenland) which are based on more than five sites. Thus, we have little knowledge of the apparent polar wander path for North America during an interval when a large apparent polar motion must have occurred. This Late Cretaceous–Eocene interval is also the time of the Laramide orogeny. Detailed knowledge of Paleocene and Eocene pole positions is essential in order to determine whether this orogenic event is reflected in the apparent polar wander path. This paper presents results of the first extensive study of the Paleocene pole position for the stable part of North America.

PALEOMAGNETIC DATA

Terrestrial deposits in the San Juan Basin of northwestern New Mexico contain well-exposed, flat-lying sedimentary rocks and well-documented vertebrate faunas for the Late Cretaceous–early Tertiary interval. Type faunas for three Paleocene Land Mammal Ages (Puercan, Torrejonian, and Tiffanian) were collected in the San Juan Basin. These faunas have been discussed by Matthew (1937) and Simpson (1935a, 1935b). We have undertaken a magnetostratigraphic study of terrestrial deposits in the San Juan Basin (Butler and others, 1977; Lindsay and others, 1978). The natural remanant magnetization (NRM) of paleomagnetic samples from the Nacimiento Formation exposed in Kutz Canyon (lat 36.55°N, long 252.08°E) was found to be particularly strong and stable. This observation suggested the possibility of analyzing the data for determination of a paleomagnetic pole position for middle Paleocene time in North America.

All fossils recovered throughout the 450-m-thick stratigraphic section from which the paleomagnetic samples for this study were collected belong to the middle Paleocene Torrejonian Land Mammal Age (Wilson, 1951; Taylor, 1977; Lindsay and others, 1978). Magnetostratigraphic data from the Kutz Canyon section show six superposed magnetozones, and these magnetozones have been correlated with magnetic anomalies 25 through 27 of the geomagnetic polarity time scale (Lindsay and others, 1978; LaBrecque and others, 1977). Thus both the paleontologic and magnetostratigraphic data require a middle Paleocene age for the Nacimiento Formation in Kutz Canyon.

A total of 158 paleomagnetic sites, with an average of three oriented block samples per site, were collected at an average stratigraphic separation of 3 m through the 450-m-thick section in Kutz Canyon. Lithologies range from dark claystones to occasional coarse channel sandstones. Paleomagnetic sampling was confined to claystones and fine siltstones. Typical be-

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havior during alternating-field (AF) demagnetization is shown in Figure 1 for one normal polarity site and one reversed polarity site. Secondary viscous components were successfully removed for most sites by AF demagnetization to a peak field of 300 Oe. The average intensity of NRM after demagnetization at 300 Oe was approximately 2 × 10^{-6} G. Thermal demagnetization studies and thermomagnetic analysis indicate that magnetite is the carrier of the NRM. The primary NRM is a depositional remanence (DRM) formed at the time of deposition or a postdepositional remanence that became blocked during consolidation (see review of DRM mechanisms by Verosub, 1977). From the 158 sites, 104 were chosen for use in the pole-position analysis on the basis of two criteria. Sites with average NRM vectors of intermediate polarity (more than 40° removed from the mean normal or mean reversed directions) were not used because these sites would not reflect the direction of the geomagnetic field in the stable normal or reversed polarity state. Also, sites with $\alpha_{95} > 40°$ after AF demagnetization at 300-Oe peak field were rejected. This criterion rejects sites whose scatter of NRM vectors indicates that they may not have accurately recorded the direction of the geomagnetic field. Mean directions for each of the sites used in this analysis are shown in Figure 2; 42 sites are of normal polarity, and 62 have reversed polarity. A mean direction for normal sites was calculated by giving unit weight to each normal site, and a mean direction for reversed sites was calculated using the same technique. These mean directions are also shown in Figure 2 along with the circles of 95% confidence.

Table 1 lists the mean directions for the normal sites and reversed sites along with the formation mean direction. The stability of the primary DRM and efficiency of AF demagnetization in removing secondary components are indicated by the observation that the mean directions of normal and reversed polarity sites are almost perfectly antiparallel. The overall mean direction was calculated by averaging antipoles of directions from reversed sites with directions from normal sites. It is noteworthy that the temporal averaging of the geomagnetic field afforded by this data set is rarely obtained in paleomagnetic studies. The mean direction calculated is the result of sampling the middle Paleocene geomagnetic field direction at 104 points in time that are roughly equally spaced over the time interval from magnetic anomaly 25 through anomaly 27.

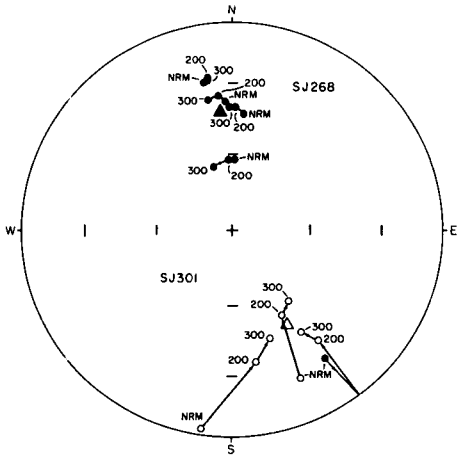


Figure 1. AF demagnetization behavior of sites SJ268 and SJ301. The equal-area stereographic projection shows directions in the lower hemisphere by solid circles; directions in the upper hemisphere are shown by open circles. Numbers by each direction indicate the peak AF field. Triangles are the site mean directions after AF demagnetization at 300-Oe peak field. α_{95} for site SJ268 is 17.2°; α_{95} for SJ301 is 14.9°.

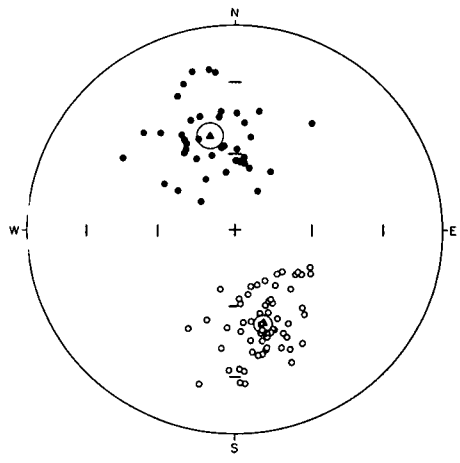


Figure 2. Equal-area stereographic projection of the site mean directions. Solid circles are directions in the lower hemisphere; open circles are directions in the upper hemisphere. Solid triangle is the mean of the 42 normal polarity sites; open triangle is the mean of the 62 reversed polarity sites. Enclosing circles are the 95% confidence circles about the mean directions.

According to the magnetic polarity time scale of LaBrecque and others (1977), this time interval was of about 4-m.y. duration from 59 to 63 m.y. B.P.

The only uncertainty about the validity of using the mean direction obtained here for determination of a pole position is the possibility of an inclination error being present in the data. Although Opdyke and Henry (1969) showed that no inclination error is present in oceanic sediments, the depositional remanence in nonbioturbated terrestrial sediments could be subject to a small inclination error. It should be noted that such an inclination error, if present, would yield a mean inclination slightly shallower than the geomagnetic field at the time of deposition, but the mean declination would not be affected. In the absence of any evidence for an inclination error, we will assume that the calculated pole position is an accurate determination of the rotation axis relative to North America in middle Paleocene time. Calculation of the paleomagnetic pole yields λ_p = 75.9°N, ϕ_p = 147.7°E with δp = 2.7° and δm = 4.1°. This pole position is illustrated in Figure 3. To our knowledge, this is the only reliable paleomagnetic pole of Paleocene age from North America (excluding Greenland).

DISCUSSION

In order to obtain supporting evidence for the pole position determined in this study, it would be desirable to compare the present result with pole positions obtained from other North American rocks of Paleocene age. The only other pole positions of similar age based on more that three sites were obtained from analyses of rocks from Greenland. Deutsch and Kristjansson (1974) and Athavale and Sharma (1975) reported results from a large collection of basalts exposed on Disko Island, West Greenland. However, the age of these rocks is somewhat uncertain and may be Late Cretaceous rather than Paleocene. Tarling (1967) reported results from Paleocene lavas at Scoresby Sound, East Greenland, but the confidence oval about the pole is rather large (α_{95} = 15°). In any case, comparison of paleomagnetic results from Greenland with other North American results is very difficult because the timing and details of opening of the Labrador Sea are not as yet well known (Talwani and Eldholm, 1977; Kristoffersen and Talwani, 1977).

TABLE 1. MEAN DIRECTIONS FOR THE NACIMIENTO FORMATION

	Inclination (°)	Declination (°)	N	R	k	α_{95} (°)
Normal polarity sites	51.7	345.2	42	39.7	17.8	5.4
Reversed polarity sites	-51.0	163.0	62	59.7	26.5	3.6
Formation mean	51.3	343.9	104	99.3	21.9	3.0

Note: N, R, k, and α_{95} are standard paleomagnetic parameters (Fisher, 1953).

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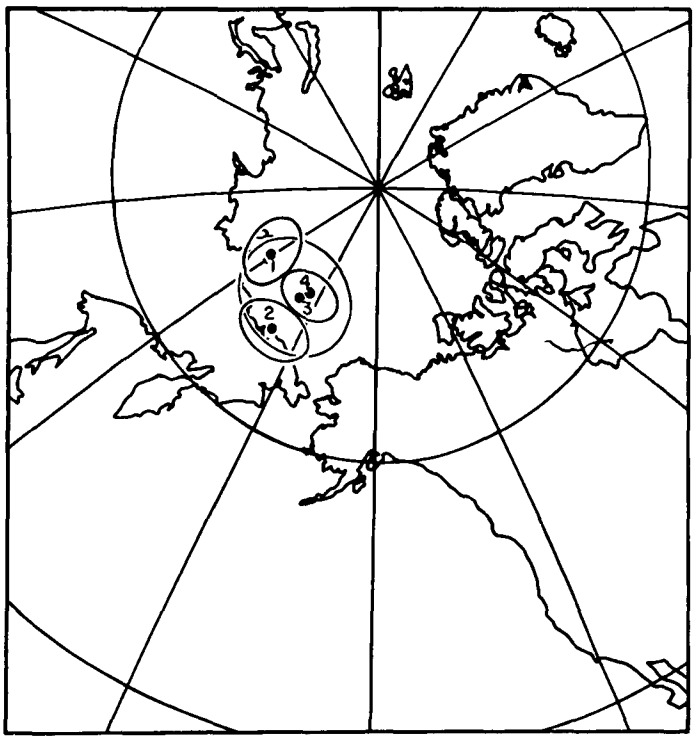


Figure 3. Pole positions from Nacimiento Formation along with three Paleocene poles from the British Tertiary igneous province. Pole 1 is middle Paleocene pole from the Nacimiento Formation (this study). Poles 2 and 3 are from Skye lavas and Vaternish dike swarm, Scotland (Wilson and others, 1972, 1974). Pole 4 is Faeroe Islands pole (Tarling, 1970). Ovals around poles are the 95% confidence ovals.

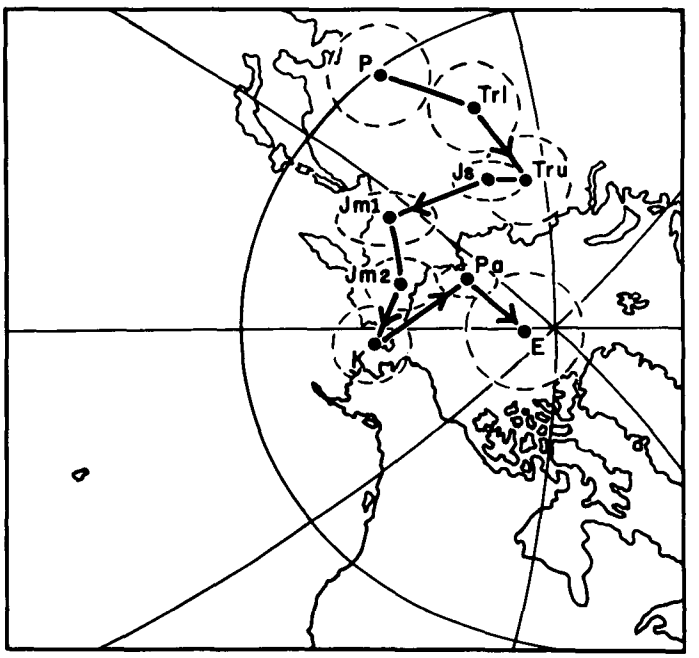


Figure 4. Revised North American apparent polar wander path. P = Permian; Trl = Lower Triassic; Tru = Upper Triassic; Js, Jm_1, Jm_2 are Upper Jurassic poles from the Summerville Formation and the lower and upper Morrison Formation, respectively (Steiner, 1978); K = Cretaceous; Pa = Paleocene (this study); E = Eocene (see text for details). Map is a stereographic projection with geographic grid lines at 45° spacings. Ovals around poles are the 95% confidence ovals.

Supporting evidence for the Paleocene pole position in this study can be obtained from analysis of paleomegnetic results from rocks of similar age from the Faeroe Islands and from Scotland. Tarling (1970) reported paleomagnetic results from 253 lavas on the Faeroe Islands. The K-Ar ages of these lavas are in between 50 and 60 m.y. (Tarling and Gale, 1968). Paleomagnetic results are also available from 87 cooling units of the Vaternish dike swarm (Wilson and others, 1974) and from 90 flows of the Skye lavas (Wilson and others, 1972) in Scotland. Most of the volcanism in the British Tertiary igneous province is thought to have occurred during Paleocene time (Mussett and others, 1972). In order to compare the pole positions obtained from these three regions on the European plate with the Paleocene pole position obtained from New Mexico, the European plate and its pole positions must be rotated toward North America about the pole of spreading of the North Atlantic. We have used the finite rotation required to close the North Atlantic north of the Azores back to anomaly-25 time (Pitman and Talwani, 1972). Pole positions before and after this rotation are given in Table 2. The pole positions and confidence ovals after rotation are illustrated in Figure 3. Considering the uncertainties in age equivalence of the poles and the rotations required for closing the North Atlantic, the agreement between the three European Paleocene poles and the pole position from the Nacimiento Formation is quite good.

It should be noted, however, that improved agreement between the Nacimiento Formation pole and the European Paleocene poles would be obtained if the Nacimiento pole were moved ~6° toward the collection locality. As an interesting speculation, we point out that a 6° increase in the mean inclination from the Nacimiento Formation would yield a pole (lat 77°N, long 173°E) that is statistically indistinguishable from the Vaternish and Faeroe poles. We do not feel that the present data set and comparison with European poles shown here should be taken as evidence for an inclination error. The present data are not adequate for such a conclusion. This problem will only be solvable when paleomagnetic results from North American Paleocene volcanic rocks become available. For the present, the comparisons of Figure 3 provide evidence that the pole position calculated in this study is a valid result for the Paleocene of North America.

The revision of the apparent polar wander path of North America necessi-

TABLE 2. PALEOCENE PALEOMAGNETIC POLES FROM EUROPEAN PLATE

Rock unit (lat, long)	Pole location before rotation (lat, long)	Pole location after rotation* (lat, long)	δp, δm	Reference
Faeroe Islands lavas (62°N, 353°E)	77.0°N, 161.0°E	75.8°N, 174.5°E	2.7°, 3.3°	Tarling (1970)
Skye lavas (57.4°N, 353.7°E)	71.5°N, 165.2°E	70.3°N, 170.8°E	2.8°, 3.8°	Wilson and others (1972)
Vaternish dikes (57.6°N, 353.4°E)	75.9°N, 160.1°E	74.9°N, 171.9°E	6.1°, 7.7°	Wilson and others (1974)

*European plate rotated toward North America. Pole of rotation: lat 63.0°N, long 157.0°E. Finite rotation angle = 14.0°. See Pitman and Talwani (1972) for details of derivation of finite rotation.

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tated by the Paleocene pole position presented here is shown in Figure 4. The Eocene pole position shown is the mean of the three reliable Eocene results available at this time (Løvlie and Opdyke, 1974; Shive and Pruss, 1977; Shive and others, 1977). This mean pole for the Eocene of North America is at lat 86.2°N, long 187.3°E with A_{95} = 8.3°. The Permian, Triassic, and Cretaceous poles of Figure 4 are from the compilation by McElhinny (1973), and the Jurassic poles are after Steiner (1975, 1978). There is substantial apparent polar wander from Permian through Early Cretaceous time, with rapid apparent polar motion during Late Jurassic time (Steiner, 1975). The rapid apparent polar wander in the Late Jurassic is followed by little or no polar motion during the Cretaceous.

Because the Paleocene pole position is based on only the present data, which could be subject to an inclination error, we do not think that the details of the early Tertiary portion of the polar wander curve in Figure 4 (that is, the dogleg in the Paleocene) should be taken too literally. However, the Paleocene pole position presented here does require that apparent polar wander between the Cretaceous pole position and the Eocene pole near the present rotation axis must have begun in the latest Cretaceous–early Paleocene interval. Thus, the latest Cretaceous–early Tertiary interval was indeed a time of rapid apparent polar wander.

Irving and Park (1972) suggested that the path of apparent polar wander relative to North America has several sharp turns that they called "hairpins." These hairpins were correlated with orogenic episodes. The youngest of their hairpins is in Cretaceous time, and Irving and Park (1972) correlated this change in apparent polar wander with a tectonic reorganization suggested by Coney (1971). The revised apparent polar wander path of Figure 4 substantiates the necessity for a hairpin turn in the Late Cretaceous–early Paleocene interval. This change in apparent polar wander relative to North America correlates in time with the Laramide orogeny and also with major changes in plate geometry as suggested by Coney (1972, 1978).

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